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Q1(a): Give one technique that programmers use to make their code easy to read and understand.

Q1(b): Programs may contain logic, syntax or runtime errors. A program has three errors. These may be logic, syntax or runtime errors. Complete the table by adding a tick (ü) for each error description to show whether the error is a logic, syntax or runtime error.

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Q1(c): Figure 1 shows a programming construct in a flowchart. Identify the name of the programming construct.

Q1(d): Open Q01d in the code editor. The program inputs two numbers, compares them and outputs a suitable message. There are three errors in the code. Amend the code to correct the errors. Save your amended code as Q01dFINISHED with the correct file extension for the programming language.

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Q1(e): A program is needed to calculate and print a multiplication table for a number that is input by the user. Open Q01e in the code editor. Write the program. You must use the structure given in the file Q01e to complete the program. Do not add any further functionality. Save your code as Q01eFINISHED with the correct file extension for the programming language.

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Q2(a): Open file Q02a in the code editor. Amend the code to complete the program. You must use the structure given in Q02a to write the program. Do not add any further functionality. Save your code as Q02aFINISHED with the correct file extension for the programming language.

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Q3(a)(i): Identify the name of a local variable.

Q3(a)(ii): Identify the name of a data structure.

Q3(a)(iii): Identify the name of a built-in subprogram.

Q3(a)(iv): Identify the name of a user-written function.

Q3(a)(v): Identify the name of a parameter.

Q3(a)(vi): The program includes validation to ensure the month number is between 1 and 12 inclusive. Complete the table by adding the type of test data that is being used to test the validation. The first type

of test has been completed for you.

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Q3(b): Explain one reason why parameters are used to pass data into subprograms.

Q3(c): A program is needed that will accept the input of a year and output whether it is a leap year or not. Open Q03c in the code editor. Write the program. You must use the structure given in Q03c. Do not add any further functionality. Save your code as Q03cFINISHED with the correct file extension for the programming language.

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Q4(a): Open Q04a in the code editor. Write a program to implement these requirements. You must use the structure given in Q04a to write the program. Do not add any further functionality. Save your code as Q04aFINISHED with the correct file extension for the programming language.

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Q5: Write the code for the program. Your program should function correctly even if the number of words in the array is changed. Save your code as Q05FINISHED with the correct file extension for the programming language.

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Q1(a): Identify the description of a variable in a computer program.

Q1(b): Identify the technique that improves the readability of code.

Q1(c): Complete the table by giving an example value for each data type.

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Q1(d): Describe one difference between the data used in boundary testing and the data used in erroneous testing.

Q1(e): A program should output the value 2 However, there is an error in the code and the actual output is 4 Name this type of error.

Q1(f): Open Q01f in the code editor. Amend the code to correct the errors. Save your amended code as Q01fFINISHED with the correct file extension for the programming language.

Q1(g): Open Q01g in the code editor. Amend the code to complete the if statement used to produce the output. You must use the structure given in Q01g to complete the program. Do not add any further functionality. Save your code as Q01gFINISHED with the correct file extension for the programming language.

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Q2(a)(i): Identify a line number where selection starts.

Q2(a)(ii): Identify the line number where iteration starts.

Q2(a)(iii): Identify a line number where a string and a number are printed on the same line.

Q2(a)(iv): Identify the name of a local variable.

Q2(a)(v): Identify the name of a parameter.

Q2(b): Open Q02b in the code editor. Write the program. You must use the structure given in Q02b to complete the program. Do not add any further functionality. Save your code as Q02bFINISHED with the correct file extension for the programming language.

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Q3: Open Q03 in the code editor. Write the program. You must use the structure given in Q03 to complete the program. Do not add any further functionality. Save your code as Q03FINISHED with the correct file extension for the programming language.

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Q5(a): Open Q05a in the code editor. Write the program. You must use the structure given in Q05a to write the program. Do not add any further functionality. Save your code as Q05aFINISHED with the correct file extension for the programming language.

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Q5(b)(i): Julia uses a bubble sort algorithm to sort the scores. Complete the table to show how the bubble sort algorithm will sort the scores.

Q5(b)(ii): Explain one reason why a bubble sort is very efficient in terms of memory usage.

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Q6: Open Q06 in the code editor. Write the code for the program. Your program should include at least two subprograms that you have written yourself. You must include comments in the code to explain the logic of your solution. Save your code as Q06FINISHED with the correct file extension for the programming language.

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Q1(b)(i): Give the contents of a comment.

Q1(b)(ii): Identify a logical operator used in this program.

Q1(b)(iii): Give the name of a global variable.

Q1(b)(iv): Give the name of a constant.

Q1(b)(v): Give the name of a parameter.

Q1(c)(i): State the purpose of a constant.

Q1(c)(ii): Give one benefit of using a constant.

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Q2(a)(i): Give two reasons to use subprogram libraries when developing code.

Q2(a)(ii): Identify which one of these must return a result.

Q2(a)(iii): State what is meant by the term local variable.

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Q2(b)(i): Complete the table to show examples of normal, boundary and erroneous numeric test data for the program.

Q2(b)(ii): Open Q02bii in the code editor. There are four errors in the code. Amend the code to correct the errors. Use this test data to help you find the errors. Save your amended code as Q02biiFINISHED with the correct file extension for the programming language.

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Q3(a): Give two characteristics of a trace table.

Q3(b): Complete the table to show the output for each set of inputs.

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Q3(c): Open Q03c in the code editor. Write a program to implement the logic in the flowchart. Do not add any further functionality. Save your code as Q03cFINISHED with the correct file extension for the programming language.

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Q5(b): There is an error in the loop between line 9 and line 16. Complete the table to give the line number with an error and a corrected line of pseudocode.

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Q5(c): Open Q05c in the code editor. Write the program. You must use the structure given in Q05c to write the program. Do not add any further functionality. Save your code as Q05cFINISHED with the correct file extension for the programming language.

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Q6: Open the file Q06 in the code editor. Write a program to: calculate the daily volume of milk produced by each breed; add this daily volume to the data structure tbl_dailyVolume; display a message informing the user what each field holds, such as breed, rating, volume per cow, count and total volume; display the data for each breed; calculate and display the total volume of milk produced each day by the herd; find the recommended breed; display the recommended breed by name. Save your amended code as Q06FINISHED with the correct file extension for the programming language.

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Q1(b): Open Q01b in the code editor. Amend the code to complete the program. Save your amended code as Q01bFINISHED with the correct file extension for the programming language.

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Q1(c)(i): Give the text of a comment used in the code.

Q1(c)(ii): Give the text of a line that creates and initialises a variable.

Q1(c)(iii): Give the keyword that starts the selection used in the code.

Q1(c)(iv): Give the logical operator used in the code.

Q1(d): Identify the data that must be stored in a record rather than an array.

Q1(e): Complete the table to show the data type for each item.

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Q2(a)(ii): Give the name for a memory location whose value can change during program execution.

Q2(b)(i): State what is meant by the term runtime error.

Q2(b)(ii): Give one example of a runtime error.

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Q2(c): Open Q02c in the code editor. There are three errors in the code. Amend the code to correct the errors. Save your amended code as Q02cFINISHED with the correct file extension for the programming language.

Q2(d): Complete the table to show two examples of erroneous, one example of normal and one example of boundary test data for the memorable word field. Each example must be different.

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Q3(a): The truth table shows the initial values for A and B. Complete the truth table to show the results of each operation.

Q3(b): Open Q03b in the code editor. Amend the code to display every other name in the array of star names. Do not add any further functionality. Save your code as Q03bFINISHED with the correct file extension for the programming language.

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Q3(c): Open Q03c in the code editor. Amend the code to ensure the messages are generated correctly. Do not add any further functionality. Save your code as Q03cFINISHED with the correct file extension for the programming language.

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Q4(a)(i): State where a global variable is created.

Q4(a)(ii): State where a local variable is accessible.

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Q4(b): Explain the reason that the algorithm in Figure 4 is more efficient than the algorithm in Figure 5.

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Q4(c): Open Q04c in the code editor. Amend the code to: complete the subprogram to construct the new key; complete the call to the subprogram. Do not add any further functionality. Save your code as Q04cFINISHED with the correct file extension for the programming language.

From june-2024 | Page 13:

Q5(a): Explain one improvement to this algorithm that will reduce the number of variables required and will enable it to work with any number of animals.

Q5(b): Complete the table to show the values for the high index, the low index and the midpoint on each pass of a binary search.

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Q5(c): Open Q05c in the code editor. Amend the code to produce the intended output. You must use the structure and variables given in Q05c to complete the program. Do not add any further functionality. Save your code as Q05cFINISHED with the correct file extension for the programming language.

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Q6: Open the file Q06 in the code editor. Write a program to produce the intended output. You must use the structure and variables given in Q06 to complete the program. You should use techniques to make your code easy to read. Save your amended code as Q06FINISHED with the correct file extension for the programming language.